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SUBJECT:	ADBMS
SECTION:	610-B

Practice 2 - Transactions & Concurrency Control

Part A: Prevent Duplicate Enrollments Using Locking

Part B: Use SELECT FOR UPDATE to Lock Student Record

Part C: Demonstrate Locking Preserving Consistency in Concurrent Transactions

CODE:

DROP TABLE IF EXISTS StudentEnrollments;

CREATE TABLE StudentEnrollments (

enrollment_id INT PRIMARY KEY,

student_name VARCHAR(100) NOT NULL,

course_id VARCHAR(10) NOT NULL,

enrollment_date DATE NOT NULL,

CONSTRAINT unique_student_course UNIQUE (student_name, course_id)

);

START TRANSACTION;

INSERT INTO StudentEnrollments VALUES (1, 'Ashish', 'CSE101', '2024-07-01');

INSERT INTO StudentEnrollments VALUES (2, 'Smaran', 'CSE102', '2024-07-01');

INSERT INTO StudentEnrollments VALUES (3, 'Vaibhav', 'CSE101', '2024-07-01');

COMMIT;

START TRANSACTION;

INSERT INTO StudentEnrollments VALUES (4, 'Ashish', 'CSE101', '2024-07-02');

ROLLBACK;

START TRANSACTION;

SELECT * FROM StudentEnrollments

WHERE student_name = 'Ashish' AND course_id = 'CSE101'

FOR UPDATE;

START TRANSACTION;

INSERT INTO StudentEnrollments VALUES (4, 'Ashish', 'CSE101', '2024-07-02');

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COMMIT;

START TRANSACTION;

SELECT * FROM StudentEnrollments

WHERE student_name = 'Ashish' AND course_id = 'CSE101'

FOR UPDATE;

START TRANSACTION;

SELECT * FROM StudentEnrollments

WHERE student_name = 'Ashish' AND course_id = 'CSE101'

FOR UPDATE;

UPDATE StudentEnrollments

SET enrollment_date = '2024-07-10'

WHERE student_name = 'Ashish' AND course_id = 'CSE101';

COMMIT;

START TRANSACTION;

UPDATE StudentEnrollments

SET enrollment_date = '2024-07-15'

WHERE student_name = 'Ashish' AND course_id = 'CSE101';

COMMIT;

SELECT * FROM StudentEnrollments

```

OUTPUT:

enrollment_id	student_name	course_id	enrollment_date
2	Smaran	CSE102	2024-07-01T00:00:00.000Z
3	Vaibhav	CSE101	2024-07-01T00:00:00.000Z
4	Ashish	CSE101	2024-07-15T00:00:00.000Z